

Scenario 1:

A large Savings and Credit Co-operative Society (SACCOS) in Dar es Salaam is migrating 500GB of sensitive member financial records to a secure cloud backup. The ICT Manager must choose between several cryptographic methods to ensure the data remains confidential during the multi-hour upload process without causing system timeouts.

Required;

- a) Define the terms Algorithm and Cryptogram in the context of this migration.
- b) Explain why a hacker intercepting the upload would only see the latter.
- c) Analyze why Symmetric Encryption (such as AES-256) is technically more suitable for this "bulk data" task compared to Asymmetric methods.
- d) If the manager chooses an extremely complex algorithm that is "technically unbreakable" but so computationally heavy that it causes the SACCOS banking app to lag and time out, evaluate if this is a successful security solution based on the Business Mission principle.

Scenario 2:

A student at the National Institute of Transport (NIT) has completed a sensitive research project on Critical Infrastructure. They need to send this project to their supervisor. To ensure security, they decide to use Asymmetric Encryption.

Required;

- a) Describe the roles of the Public Key and the Private Key in this transaction.
- b) Which key should the student use to encrypt the file so that only the supervisor can read it?
- c) The student also wants to provide a Digital Signature. Analyze the step-by-step process of how hashing and asymmetric encryption combine to create this signature.
- d) Explain how this Digital Signature provides Non-repudiation, and why this is legally important for academic or official institutional documents.

Scenario 3:

A Tanzanian mobile banking startup is designing its security architecture. The lead engineer insists that user passwords should never be stored in plain text. Instead, they propose using Cryptographic Hashing.

Required;

- a) Analyze the characteristics of a Hash Function (Digital Fingerprint), specifically focusing on its "one-way" property and "avalanche effect".
- b) The team is debating between using MD5 and SHA-256. Analyze the concept of a Collision and explain why MD5 is no longer recommended for high-security Tanzanian financial systems.
- c) Define a Birthday Attack and explain how a cybersecurity expert uses the "Work Factor" to determine if a hash output size is large enough to resist this attack.

Scenario 4:

The National Institute of Transport (NIT) is implementing a Public Key Infrastructure (PKI) to secure their online portals and internal communications. They are currently managing various Digital Certificates, including Wildcard Certificates for their subdomains.

Required;

- a) Differentiate between a Digital Signature and a Digital Certificate.
- b) How does the certificate act as an "ID Card" for the NIT's server?
- c) Explain the role of the Certificate Authority (CA) and the Registration Authority (RA) in this framework.
- d) Why is a third-party CA certificate generally more trusted than a self-signed certificate for public-facing services?
- e) Describe the TLS Handshake process.

- f) How do asymmetric and symmetric encryption work together in a "Hybrid Cryptosystem" during this handshake to ensure both security and speed?

Scenario 5:

A water treatment plant in a rural Tanzanian region uses industrial controllers (OT) that are decades old. These legacy devices lack the processing power for modern, heavy encryption but are now being connected to the main IT network.

Required;

- a) Analyze the constraints of Legacy OT in this scenario.
- b) Why might a standard 256-bit AES block cipher be impractical for these specific industrial controllers?
- c) Compare the use of Stream Ciphers versus Block Ciphers in this environment. Which is typically better for real-time traffic or devices with very limited hardware resources?
- d) Apply Kerckhoffs's Principle to this plant's security.
- e) If an attacker discovers the specific cryptographic algorithm being used (e.g., DES or Blowfish), explain why the system should still remain secure according to Kerckhoffs's Principle.